



Mahindra University, Hyderabad
École Centrale School of Engineering
Minor II Examination

Program: B. Tech. Branch: CM Year: II Semester: II
Subject: Number Theory & Cryptography (MA 2209)

Date: 17/04/2026
Time Duration: 1.5 Hours

Start Time: 2.00 PM
Max. Marks: 18

Instructions:

1. There are 4 questions, all of which are compulsory.
2. Justify your answer wherever required.

1. (a) Evaluate the Jacobi symbol $\left(\frac{610}{987}\right)$. [2]
(b) Is 3 a quadratic residue modulo 131? Justify your answer. [2]
(c) State Pollard's $p - 1$ factorization algorithm. [2]

2. Given that the order of 374 modulo 433 is 27, solve the following discrete logarithm problem using the Pohlig-Hellman algorithm: [4]

$$374^x \equiv 335 \pmod{433}.$$

3. State and prove Euler's Criterion for quadratic residues modulo p , where p is a prime. [5]

OR

Let p be an odd prime and a be an integer not divisible by p . Prove that if a is a quadratic residue modulo p then it is a quadratic residue modulo p^k for all positive integers k .

OR

Find a solution to the congruence $x^2 \equiv 5 \pmod{121}$ using a solution modulo 11.

4. Let p be a prime and $g \in \mathbb{F}_p^*$ be a primitive element. Let $n = 1 + \lfloor \sqrt{p-1} \rfloor$. For any non-zero $h \in \mathbb{F}_p^*$, show that the following two sets have at least one element in common:

$$\{e, g, g^2, \dots, g^n\} \text{ and,}$$

$$\{h, hg^{-n}, hg^{-2n}, hg^{-3n}, \dots, hg^{-n^2}\}.$$

[3]
